Speech Emotion Recognition Under Speaker-Independent Evaluation

A Paralinguistic Feature-Based Study on RAVDESS

Dataset	Audio Samples	Speakers	Main Accuracy
RAVDESS	**2,880**	**24**	**48.33%**

Abstract

This report summarizes a research-oriented notebook on speech emotion recognition using the RAVDESS dataset. The study investigates whether emotional states can be inferred from speech using acoustic and paralinguistic descriptors, while placing particular emphasis on **generalization across unseen speakers**. The main methodological finding is that strong performance under a standard random split does not translate directly to a speaker-independent setting. In the most realistic evaluation protocol, the best model reaches **48.33% accuracy**, compared with **92.36%** under the easier standard split.

Main research question: Can emotion recognition models generalize to unseen speakers, or do they mainly learn speaker-specific patterns?

1. Context and Research Motivation

Speech contains more than lexical information. In addition to words, spoken utterances convey paralinguistic cues such as tone, rhythm, speaking rate, intensity, and emotional coloring. These cues are central to human communication because they influence how an utterance is interpreted, even when the literal linguistic content remains unchanged.

From a machine learning perspective, this makes speech emotion recognition both interesting and difficult. Emotional categories are not perfectly separable, and speaker identity often introduces strong acoustic variation. As a result, high performance may be misleading if the evaluation protocol is not sufficiently strict.

2. Objective of the Study

The objective of this project is to build a reproducible speech emotion recognition pipeline and to evaluate it under conditions that better reflect realistic deployment. More specifically, the notebook aims to:

- extract informative acoustic features from speech signals,
- train fast and interpretable baseline models,
- compare standard and speaker-independent evaluation protocols,
- analyze not only performance, but also confusion patterns and model limitations.

The scientific importance of the study lies in the distinction between **optimistic random-split performance** and **speaker-independent generalization**, which is the main criterion retained for interpretation.

3. Dataset Description

The experiments are conducted on the speech portion of the **RAVDESS** dataset (Ryerson Audio-Visual Database of Emotional Speech and Song), a standard benchmark for speech emotion recognition research.

Property	Value
Total audio samples	2,880
Number of speakers	24
Emotion classes	8 (angry, calm, disgust, fearful, happy, neutral, sad, surprised)
Recording conditions	Controlled studio-like setting

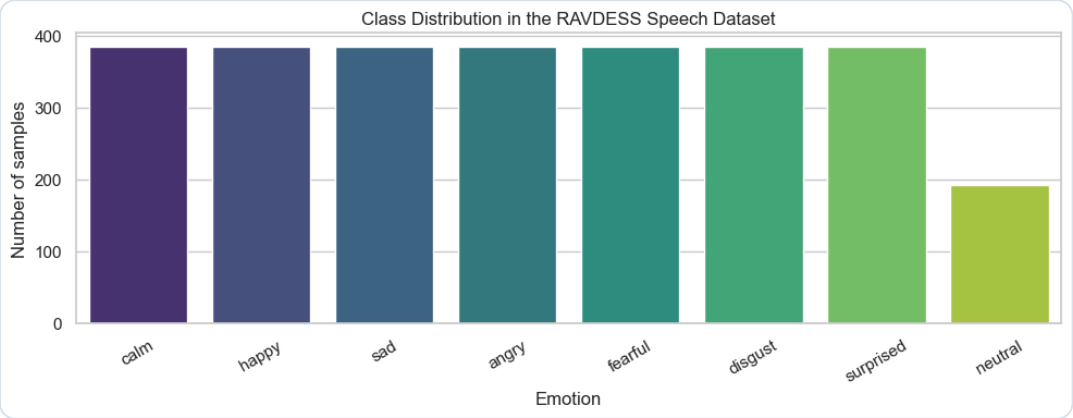

Figure 1. Distribution of the eight emotion classes in the RAVDESS speech subset. Most classes are close to 384 samples, while the `neutral` class is smaller (about 192 samples), which motivates reporting balanced accuracy.

The dataset is appropriate for this study because it combines controlled conditions with speaker variability. This allows the notebook to assess whether models truly learn emotion-related acoustic structure or simply benefit from repeated exposure to the same voices.

4. Processing Pipeline and Feature Engineering

The notebook follows a complete processing pipeline, starting from raw `.wav` files and ending with model evaluation and interpretation. Each audio file is scanned, matched with its label, and transformed into a compact vector representation suitable for supervised learning.

RAVDESS Audio Input
2,880 labeled utterances from 24 speakers

→

Inspection & Metadata
file parsing, emotion labels, and actor identifiers

→

Audio Preprocessing
signal loading and standard per-utterance preparation

Feature Extraction
MFCC, ΔMFCC, chroma, RMS, spectral centroid, and ZCR

→

Structured Feature Matrix
2,880 samples × 82 features

→

Modeling & Evaluation
RandomForest / MLP with standard and actor-disjoint testing

The feature set was designed to capture the main paralinguistic properties of speech while remaining interpretable and computationally efficient.

These descriptors are directly linked to how emotion is expressed acoustically through timbre, energy, articulation, and temporal dynamics, which explains their relevance for the classification task.

5. Experimental Design and Model Comparison

A central methodological aspect of the notebook is the comparison between two evaluation strategies: a standard stratified sample-level split and a stricter actor-disjoint group split.

Approach	Train Samples	Test Samples	Train Actors	Test Actors	Shared Actors
Stratified sample split	2,304	576	24	24	24
Actor-disjoint group split	2,280	600	19	5	0

The first protocol is useful as a quick baseline, but it allows the same speakers to appear in both train and test sets. The second protocol is methodologically stronger because it evaluates the model on **unseen speakers**, thereby providing a more realistic estimate of generalization.

Two baseline models are compared in the notebook:

- **RandomForest**, used as a strong classical baseline;
- **MLP**, used as a more flexible nonlinear model.

This comparison is important because the project is not limited to “which model performs best,” but also examines which model remains more robust when the evaluation becomes more realistic.

6. Quantitative Results

The notebook shows a substantial contrast between the two evaluation settings. Performance is very high under the standard split, but drops markedly under the speaker-independent protocol.

Protocol	Model	Train Accuracy	Test Accuracy	Balanced Accuracy	Macro F1
Stratified sample split	RandomForest	1.0000	0.9236	0.9285	0.9261
Stratified sample split	MLP	0.9883	0.9149	0.9187	0.9176
Actor-disjoint group split	MLP	0.9939	**0.4833**	**0.4719**	**0.4742**
Actor-disjoint group split	RandomForest	1.0000	0.4367	0.4188	0.4153

The most important numerical finding is the drop from **92.36%** to **48.33%**, corresponding to an absolute decrease of about **44.03 percentage points** once speaker overlap is removed.

Even under the stricter actor-disjoint evaluation, the model remains clearly above the random baseline of **12.5%** for eight classes. This means that the acoustic feature set does contain meaningful emotion-related information, but that the task remains challenging when speaker generalization is required.

7. Interpretation and Error Analysis

The central interpretation of the results is that the main limitation is **overfitting to speaker-specific patterns** rather than a failure to detect any emotional information at all. This conclusion is supported by the large generalization gap between training and test performance.

Under the main actor-disjoint protocol, the selected MLP reaches **99.39%** training accuracy but only **48.33%** test accuracy, yielding a generalization gap of approximately **51.06 percentage points**.

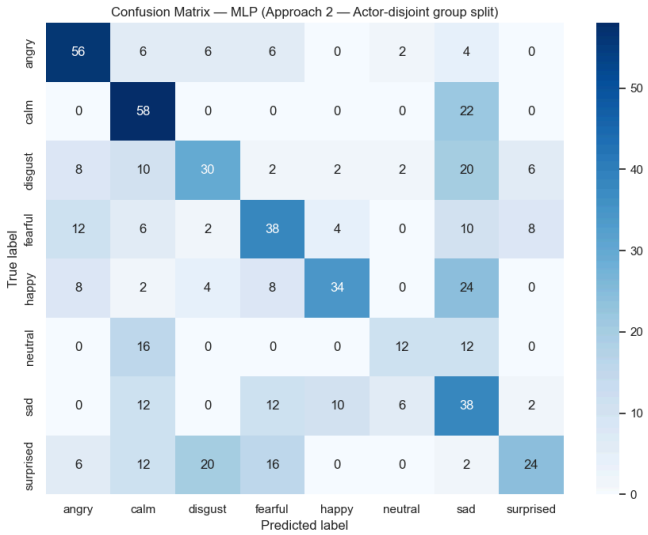

Figure 2. Confusion matrix for the MLP under the speaker-independent evaluation setting. The error structure is informative and not random.

8. Detailed Error Analysis

The class-wise report shows notable differences across emotions. The class `angry` is the strongest overall with **precision = 0.622**, **recall = 0.700**, and **F1 = 0.659**. The class `calm` reaches the highest recall at **0.725**, while `happy` attains the highest precision at **0.680**. By contrast, `sad`, `neutral`, and `surprised` remain harder to classify consistently.

The most frequent confusions include `happy → sad` (24 cases), `calm → sad` (22 cases), `disgust → sad` (20 cases), and `surprised → disgust` (20 cases). These patterns suggest that some low-energy or acoustically overlapping emotional states are collapsed into the same decision region by the classifier.

9. Conclusion and Future Work

This project confirms that speech emotion recognition is feasible with classical acoustic features, but also shows that **evaluation protocol strongly influences the conclusions**. High scores obtained under random train/test splits should not be interpreted as evidence of robust real-world performance.

The most reliable conclusion of the notebook is therefore methodological as much as predictive: when the system is evaluated on voices never seen during training, the task remains substantially more difficult, and the apparent performance drops accordingly.

Future directions

- testing end-to-end neural architectures such as CNNs or transformer-based models,
- learning more speaker-invariant representations,
- extending the system to a true multimodal setting using independent text or visual modalities.